

Model Performance Report: PhishNet.AI

1. Executive Summary

This report details the performance of two Support Vector Classifier (SVC) models trained to detect spam from the [SpamAssassin email dataset](#). The primary features were derived from the email body using TF-IDF vectorization. The model trained on the undersampled data is recommended for general use due to its higher ROC AUC score, which indicates a better ability to distinguish between spam and non-spam, and its improved recall for the minority spam class.

2. Model 1: Trained on Original (Imbalanced) Data

This model was trained on the full, class-imbalanced training set.

Key Performance Metrics:

- Accuracy: 0.976
- ROC AUC Score: 0.9586

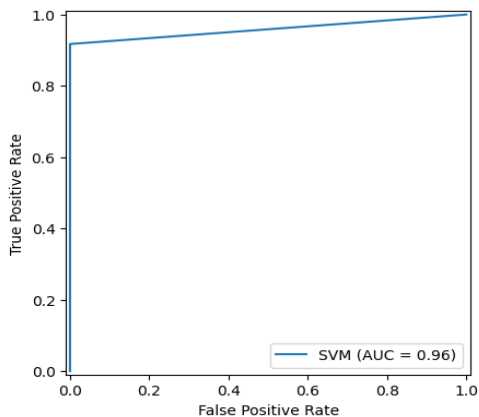


Figure 1: Initial ROC graph

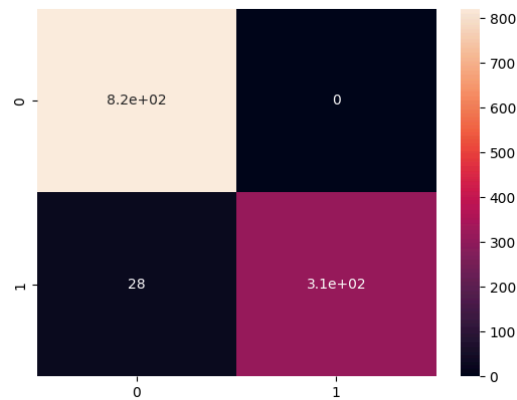


Figure 2: Initial Confusion Matrix

3. Model 2: Trained on Undersampled Data (Final model used)

This model was trained after using [RandomUnderSampler](#) to balance the training data, resulting in an equal number of spam and non-spam examples.

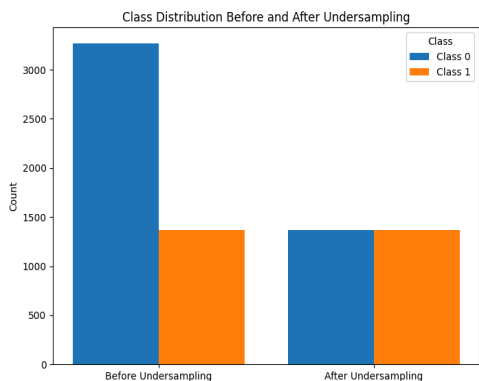


Figure 3: Class-distribution in model 1 vs 2

Key Performance Metrics:

- ROC AUC Score: 0.9746790967862833
 - The ROC score improved after undersampling, hence this model was chosen as the final. Below is the confusion matrix for the final model:

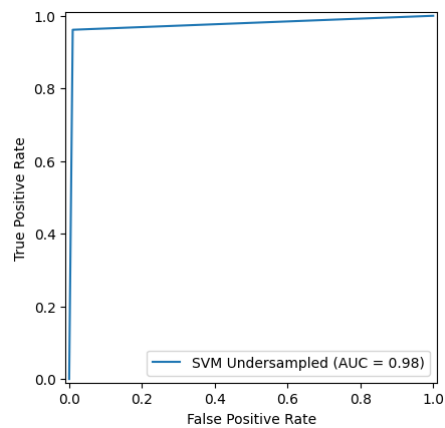


Figure 4: Final ROC graph

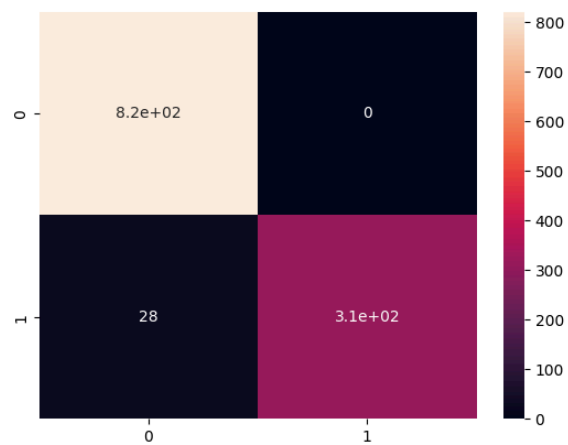


Figure 5: Confusion matrix for final model

Table 1: Final Classification report				
Category	Precision	Recall	F1-Score	Support
0	0.98	0.99	0.99 ▾	821 ▾
1	0.97	0.96	0.97 ▾	338 ▾
Accuracy			0.98 ▾	1159 ▾
Macro Avg	0.98	0.97	0.98 ▾	1159 ▾
Weighted Avg	0.98	0.98	0.98 ▾	1159 ▾

4. Final Test & Conclusion

The final test on a sample non-spam email demonstrates the model's practical application.

- Input String: "It was great connecting with you..."
- Prediction: Not Spam
- Confidence (Probability of being Spam): 9.62%

This result is excellent, as the model correctly identified the legitimate email with high certainty (a corresponding 90.38% confidence of it being "Not Spam"). The models demonstrate strong predictive capabilities suitable for real-world application.